# ComicCraft - AI Comic Story Creator using  Gemini Models

## Scenarios:

Scenario 1 : A user enters a prompt describing their comic story, such as “A brave fox exploring an enchanted forest.” The system uses this input to generate a complete comic strip, including multiple panels with accompanying images and text. The user also specifies preferences like the main character’s name, the setting (e.g., forest), tone (e.g., dramatic), and art style (e.g., anime). The system combines **Gemini 1.5 Pro** for story generation and **Stable Diffusion** for image creation to produce a visually immersive, personalized comic strip. This personalized approach enhances user satisfaction by generating a cohesive comic story that fits the user’s creative vision.

Scenario 2: A user wants a light-hearted, cartoonish feel for their comic. On the input form, they select “funny” as the tone and “comic book” as the art style (instead of the default or their previous choices). When they submit, the system regenerates the entire pipeline—outline, story, and images—this time with prompts and model instructions tailored to produce a more humorous narrative and a classic comic-book look. This lets users iterate on and customize the mood and visual style of their comic until it matches their creative vision.

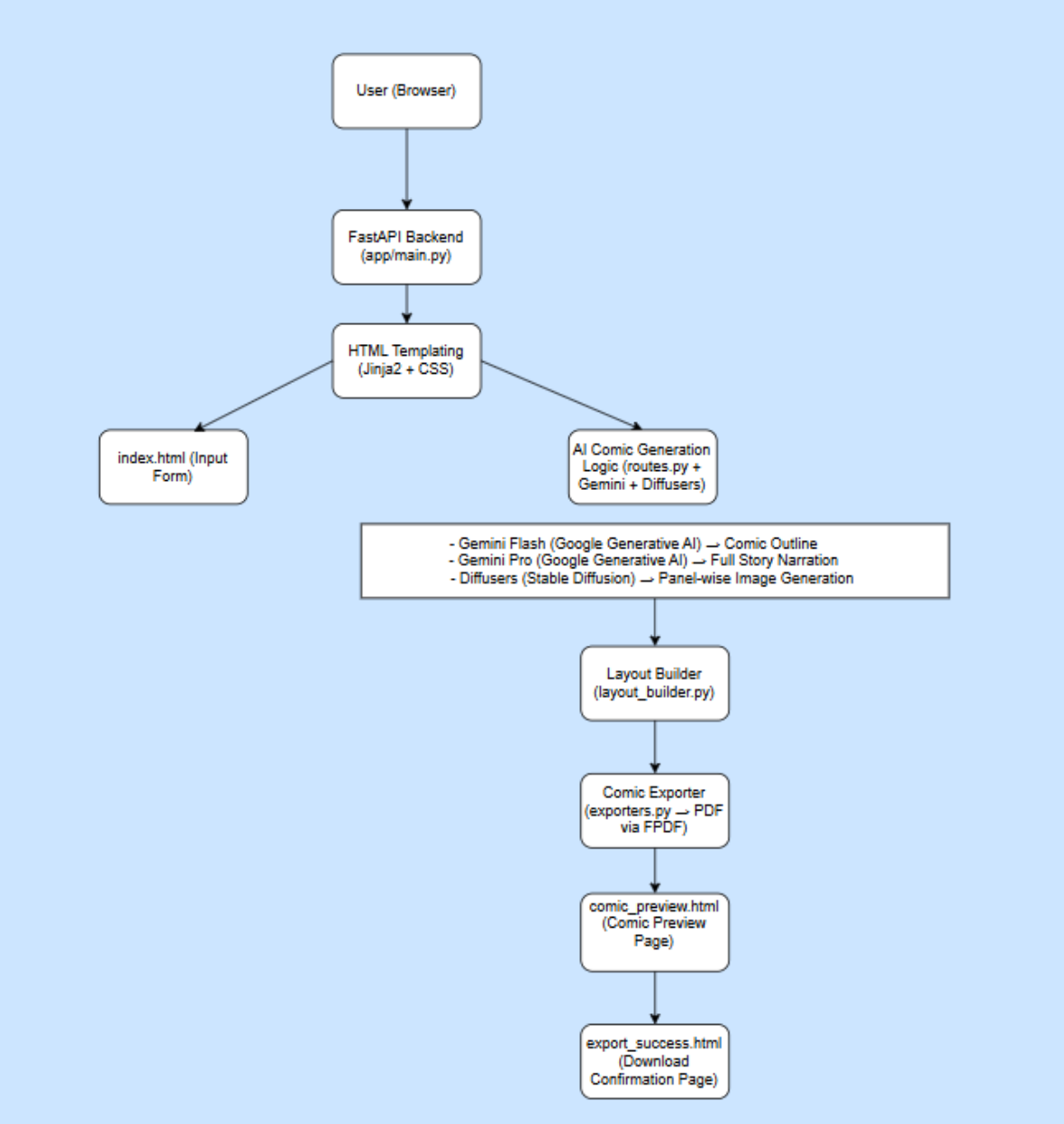
Scenario 3: After reviewing the on-screen preview of their comic, the user decides to download it. The system then:

1. **Layout Binding:** Uses layout_builder.py to assemble each panel’s title, image, and text into a structured layout.

2. **PDF Generation:** Calls exporters.py to compile the comic into a **PDF** file using **FPDF**, placing images and narrative text onto consecutive pages and saving the PDF with a timestamped filename.

3. **Export Success:** The user is redirected to an export success page, confirming the successful download. This feature ensures users can save, print, or share their comic in a professional PDF format.

## Technical Architecture:


User (Browser)
FastAPI Backend (app/main.py)
HTML Templating (Jinja2 + CSS)
index.html (Input Form)
AI Comic Generation Logic (routes.py + Gemini + Diffusers)
- Gemini Flash (Google Generative AI) → Comic Outline
- Gemini Pro (Google Generative AI) → Full Story Narration
- Diffusers (Stable Diffusion) → Panel-wise Image Generation
Layout Builder (layout_builder.py)
Comic Exporter (exporters.py → PDF via FPDF)
comic_preview.html (Comic Preview Page)
export_success.html (Download Confirmation Page)

**Pre - requisites:**

1. **FastAPI Framework Knowledge:** FastAPI Documentation
2. **Uvicorn ASGI Server:** Uvicorn Docs
3. **Python Programming Proficiency:** Python Official Docs
4. **HTML, CSS, and Jinja2 Template Skills:** W3Schools HTML/CSS/Jinja2 Tutorials
5. **Google Gemini API Familiarity:** Google Generative AI Documentation
6. **Hugging Face Diffusers & Stable Diffusion:** Hugging Face Diffusers Documentation
7. **PyTorch Basics:** PyTorch Documentation
8. **PDF Generation with FPDF:** FPDF Documentation
9. **Version Control with Git:**Git Documentation
10. **Python Package & Environment Setup (pip, virtualenv, or conda):** Virtualenv Guide